

Homework 2: Powders 101 — The Workflow and the Chemistry

Paper track. A merge of the former Homework 2 (Workflow) + Homework 3 (Chemistry), matching Session 2's merge. Pairs with Home Lab 2 and Home Lab 3. Do this on your own.

Part A: Identify from results (2 pts each)

For each unknown, give the **name and formula**.

#	Look	Water	pH	+ Acid	+ Iodine	Heat	Identity + formula
1	fine white powder	stays cloudy	~9	strong fizz	no change	just sits	
2	cubic crystals	dissolves fully	7	no fizz	no change	crackles, no melt	
3	fine white powder	"oobleck"	7	no fizz	blue-black	chars	
4	white powder	dissolves fully	~9	strong fizz	no change	just sits	
5	glassy grains	dissolves fully	7	no fizz	no change	melts, caramel smell	

Part B: Read the key (5 pts)

- | | | |
|-------------------------------------|-------------------|------------|
| 1. Fizzes strongly with acid? | yes → 2 | no → 3 |
| 2. Dissolves fully in water? | yes → BAKING SODA | no → CHALK |
| 3. Blue-black with iodine? | yes → CORNSTARCH | no → 4 |
| 4. Dissolves fully in water? | no → GYPSUM | yes → 5 |
| 5. Cubic crystals, no melt on heat? | yes → TABLE SALT | no → SUGAR |

Which numbered steps do you pass through, and what's the answer, for:

1. A powder that fizzes with acid and leaves cloudy undissolved residue.
2. A powder that doesn't fizz, is iodine-negative, dissolves fully, and melts to a brown syrup on heat.
3. A powder that doesn't fizz and turns iodine blue-black.

Part C: Build a key (4 pts)

You have four unknowns and know the list is: **Epsom salt, citric acid, corn starch, table salt**. Write a dichotomous key (first question, branches, down to all four). *Hint: which one test cleanly removes corn starch? Which removes citric acid?*

Part D: Mixture logic (2 pts)

A sample half-dissolves in water. The dissolved part is neutral and iodine- negative. The undissolved part turns iodine blue-black. Is this one powder or a mixture? Name the component(s).

Part E: Answer format (2 pts)

Rewrite this as a full scoring answer: "Sample 4 is baking soda." (Assume the suspect who bakes is "Suspect B.")

Part F: Formula recall (1 pt each)

Write the formula from memory:

1. Baking soda
2. Washing soda
3. Calcium carbonate
4. Sucrose
2. Starch
6. Table salt
7. Gypsum
8. Epsom salt
9. Citric acid
3. Vitamin C
11. Sodium acetate
12. Sand

Part G: Why the tests work (short answer)

1. **(4 pts)** Explain why and what makes corn starch turn black in iodine. (*Aim for four scoring points: the helix, the iodine going inside, the charge-transfer complex, the light absorption / color.*)
2. **(2 pts)** Is the iodine–starch test a chemical reaction? What is the correct term for what happens?
3. **(2 pts)** Amylose gives a blue-black color; amylopectin gives red-brown. Why the difference?
4. **(2 pts)** A powder turns iodine **colorless**. What is it, and why does that happen? How is this different from the starch result?
5. **(2 pts)** Why does chalk fizz more slowly with acid than baking soda does, even though both are carbonates?

Part H: Distinguish (2 pts each)

For each pair, name the single best test to tell them apart and the result for each:

1. Baking soda vs. baking powder
2. Table salt vs. citric acid
3. Corn starch vs. calcium carbonate
4. Granulated sugar vs. table salt

Part I: Benedict's (2 pts)

A question says "Benedict's solution turned brick-red." What does that mean? Would you expect table sugar to do this?

Answer Key

Part A: 1. Calcium carbonate / chalk — CaCO_3 . 2. Table salt — NaCl . 3. Corn starch — $(\text{C}_6\text{H}_{10}\text{O}_5)_n$. 4. Baking soda — NaHCO_3 . 5. Sucrose (sugar) — $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.

Part B: 1. Steps 1→2 → **chalk** (fizzes, doesn't dissolve). 2. Steps 1→3→4→5 → **sugar**. 3. Steps 1→3 → **cornstarch**.

Part C: One good key:

1. Turns iodine blue-black? yes → CORN STARCH no → 2
2. Water solution is acidic ($\text{pH} < 5$)? yes → CITRIC ACID no → 3
3. Crystals cubic (vs. needle-like), not bitter? yes → TABLE SALT no → EPSOM SALT

Part D: A mixture — table salt (soluble, neutral, iodine-negative) + corn starch (insoluble, iodine-positive).

Part E: "Sample 4: baking soda (sodium bicarbonate) — NaHCO_3 — implicates Suspect B, who bakes."

Part F: 1. NaHCO_3 2. Na_2CO_3 3. CaCO_3 4. $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ 5. $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ 6. NaCl 7. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ 8. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ 9. $\text{C}_6\text{H}_8\text{O}_7$ 10. $\text{C}_6\text{H}_8\text{O}_6$ 11. $\text{NaC}_2\text{H}_3\text{O}_2$ 12. SiO_2

Part G:

1. Starch contains **amylose**, whose molecules coil into a **helix** (1); **iodine, as triiodide ions, slides inside** the helix (1); the trapped iodine and starch form a **charge-transfer complex** (1) that **absorbs visible light**, making it look blue-black (1).
2. No — it's an **inclusion complex** (a clathrate). The iodine is physically trapped, not chemically changed.
3. Amylose is a long straight chain that can hold a long chain of iodine; amylopectin is branched, and its short branches can't fit a long iodine chain, so it only weakly complexes → red-brown.
4. Vitamin C (ascorbic acid) — it is a **reducing agent** and chemically converts the iodine to colorless iodide, so there's no iodine left to give a color. Unlike the starch test, this **is** a real chemical reaction.
5. Chalk (calcium carbonate) is **insoluble**, so the acid can only reach the surface of each grain; baking soda dissolves, so the whole amount reacts at once.

Part H:

1. **Iodine** — baking powder turns blue-black (it contains corn starch); baking soda doesn't.
2. **pH** — citric acid is acidic ($\text{pH} \sim 2$); salt is neutral ($\text{pH} 7$). (Or taste-safe note: never taste unknowns.)
3. **Water solubility** — corn starch stays a cloudy suspension / "oobleck"; calcium carbonate also stays cloudy but **fizzes with acid** and corn starch doesn't, and corn starch is iodine-positive. Best single test: iodine (starch +) or acid (carbonate fizzes).
4. **Heat** — sugar melts and caramelizes; salt crackles and does not melt. (Also: crystal shape — salt is cubic.)

Part I: A reducing sugar (glucose, fructose, maltose, lactose) is present. No — table sugar (sucrose) is a non-reducing sugar and stays blue unless it is first broken down.